(Prim's Algorithm) MST- Prim (G, w, r) — r: root vertex

1. For each $u \in V(G)$
2. $\quad key[u] \leftarrow \infty$
3. $\quad \pi[u] \leftarrow Nil$
4. $key[r] \leftarrow 0$
5. $Q \leftarrow V(G)$

6. while $Q \neq \phi$

7. $u \leftarrow$ Extract-Min(Q)

8. For each $v \in Adj[u]$

9. If $v \in Q$ and $w(u,v) < key[v]$

10. then $\pi[v] \leftarrow u$

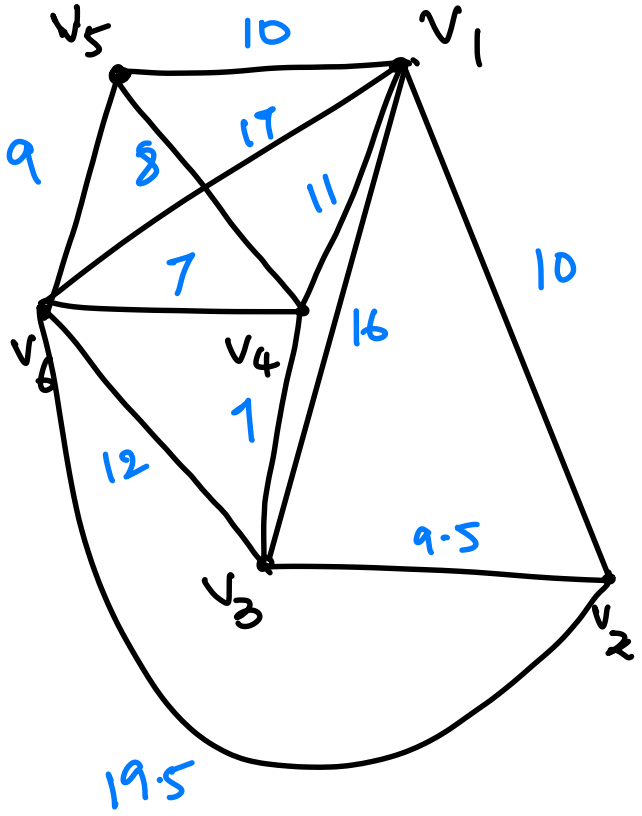
11. $key[v] \leftarrow w(u,v)$

Notations:

$\pi[v]$ ← Parent of v in the tree

key[v] ← minimum weight of any edge connecting v to a vertex in the tree.

$G =$



	v_1	v_2	v_3	v_4	v_5	v_6
Q						
key	$\not{\infty}_0$	∞	∞	∞	∞	∞
π	Nil	Nil	Nil	Nil	Nil	Nil

Let $r = v_1$

$key[r] = 0$

Tracking the edges.

1. v_1v_2 – 10
2. v_2v_3 – 9.5
3. v_3v_4 – 7
4. v_4v_6 – 7
5. v_4v_5 – 8

41.5

While $Q \neq \phi$:

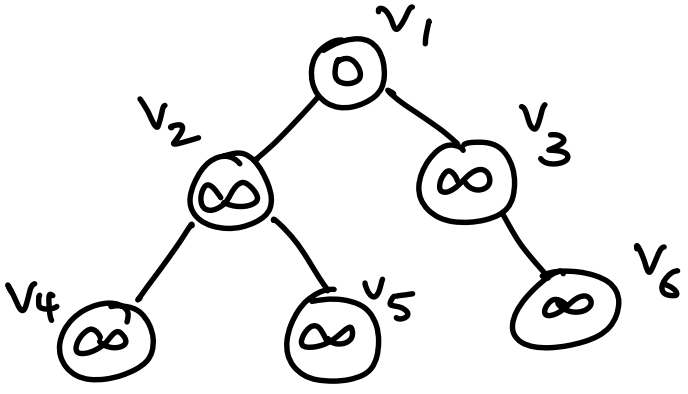
6. while $Q \neq \phi$
7. $u \leftarrow$ Extract-Min(Q)
8. For each $v \in Adj[u]$
9. If $v \in Q$ and $w(u,v) < key[v]$
10. then $\pi[v] \leftarrow u$
11. $key[v] \leftarrow w(u,v)$

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$u \leftarrow$ Extract-Min(Q)

I) Make Q a valid heap

(Current)



It is a valid Heap

II) Apply extract Min Q

Extract-Min(Q)

1. min $\leftarrow$ A[1]
2. A[1] $\leftarrow$ A[heapsize[A]]
3. heapsize[A] $\leftarrow$ heapsize[A]-1
4. Min-Heapify(A, 1)
6. return min

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Q

	v_1	v_2	v_3	v_4	v_5	v_6	
∞	$\emptyset$	∞	∞	∞	∞	$\emptyset$	$0 = A$
	Nil	Nil	Nil	Nil	Nil	Nil	

$min \leftarrow 0$

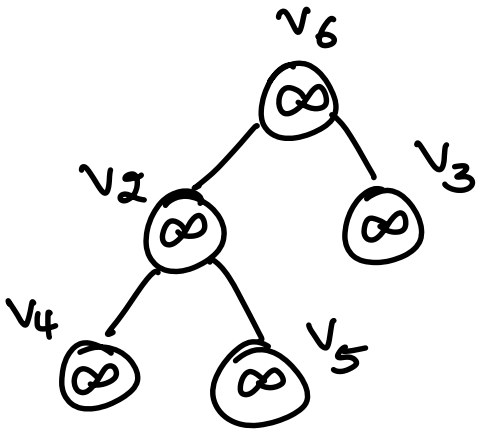
$A[1] \leftarrow A[heapsize[A]] = A[6] = \infty$

$heapsize[A] \leftarrow 6 - 1 = 5$

$\Rightarrow$ $A =$

v_6	v_2	v_3	v_4	v_5
∞	∞	∞	∞	∞

Apply Min_Heapify (A, 1)



Valid Heap.

	v_6	v_2	v_3	v_4	v_5
Q	∞	∞	∞	∞	∞
	Nil	Nil	Nil	Nil	Nil

$u \leftarrow v_1$.

$Adj[v_1] = \{ v_2, v_3, v_4, v_5, v_6 \}$ (10, 16, 11, 10, 17)

$v_2 \in Q$, $w(v_1, v_2) = 10 < \infty$

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$\pi[v_2] \leftarrow v_1$, $key \leftarrow 10$

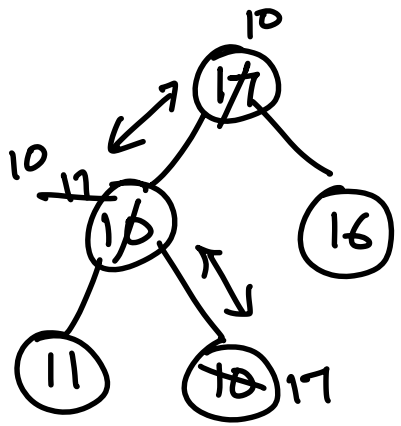
$v_3 \in Q$, $\omega(v_1, v_3) = 16 < \infty$ update

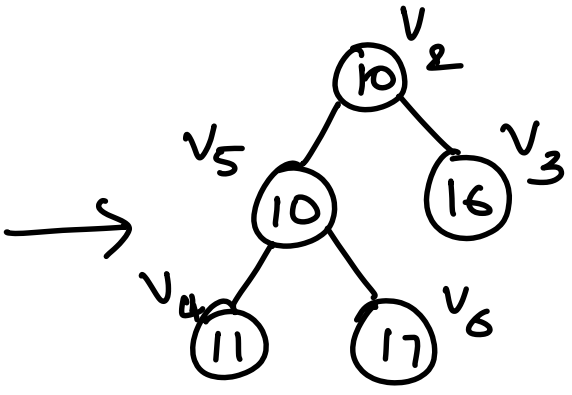
$v_4 \in Q$, $\omega(v_1, v_4) = 11 < \infty$ update

$v_5 \in Q$, $\omega(v_1, v_5) = 10 < \infty$ update

$v_6 \in Q$, $\omega(v_1, v_6) = 17 < \infty$ update

	v_6	v_2	v_3	v_4	v_5
Q	17	10	16	11	10
	v_1	v_1	v_1	v_1	v_1





Make a Valid heap before extract min

(Min_Heapify)

$i = \left\lfloor \frac{heapsize}{2} \right\rfloor$

$i = 2 \to 1.$

Valid Heap.

$i=2$ ✓

$10 < 11$

$10 \leq 10$

$i = 1$

$17 \nleq 10$

$10 < 16$

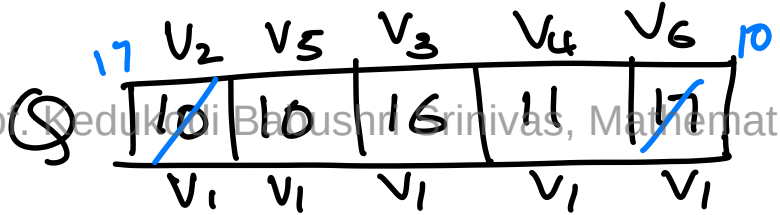
$10 \leftrightarrow 17$

$i=2$

$17 \nleq 11$

$11 \nleq 10$

$10 \leftrightarrow 17$



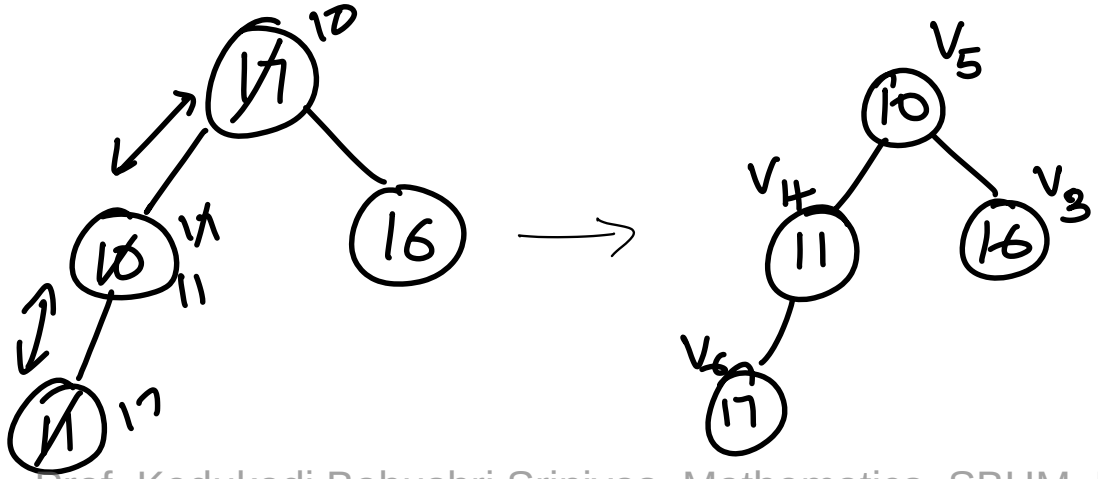
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Apply Extract-Min(Q).

min ← 10

	v_6	v_5	v_3	v_4
Q	17	10	16	11
	v_1	v_1	v_1	v_1

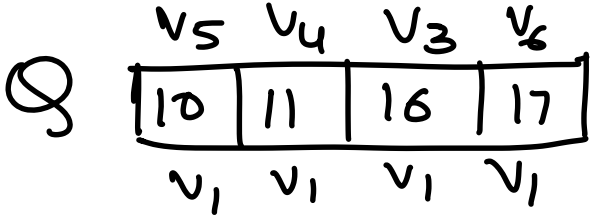
Make Q a valid heap



$i = \lfloor \frac{n}{2} \rfloor = 2$

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$i=2$: $10 < 11$. $i=1$ $17 \not< 10$ $10 < 16$ $10 \leftrightarrow 17$ $i=2$ $17 \not< 11$, $17 \leftrightarrow 11$



$u \leftarrow v_2$

$\text{Adj}[v_2] = \{ v_3, v_6, v_1 \}$ (9.5, 19.5, 10)

$v_3 \in Q$ $\quad w(v_2, v_3) = 9.5 < 16$ update

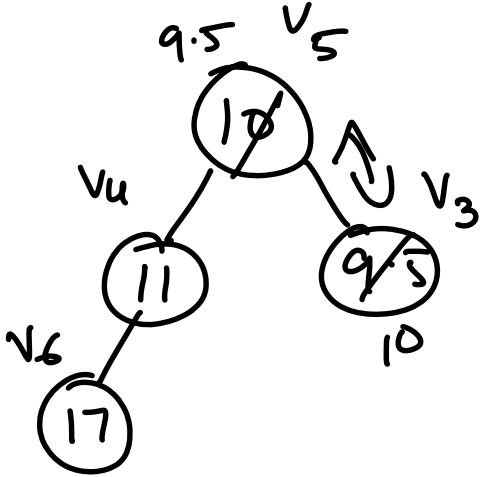
$v_6 \in Q$ $\quad w(v_2, v_6) = 19.5 \not< 17$ No change

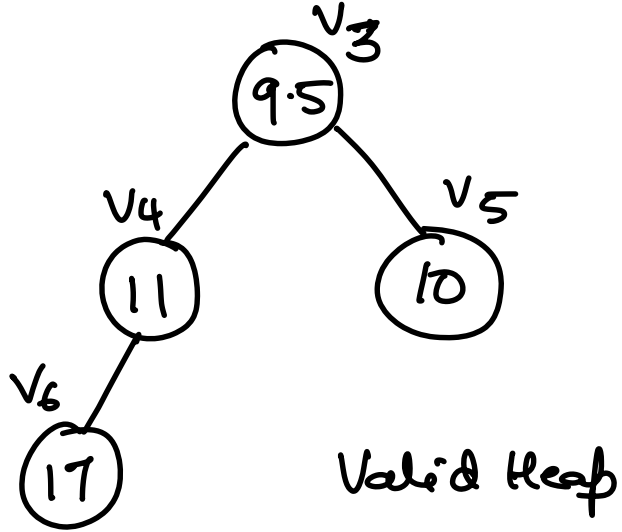
$v_1 \notin Q$.

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Q

v_5	v_4	v_3	v_6
10	11	9.5	17
v_1	v_1	v_2	v_1





Valid Heap

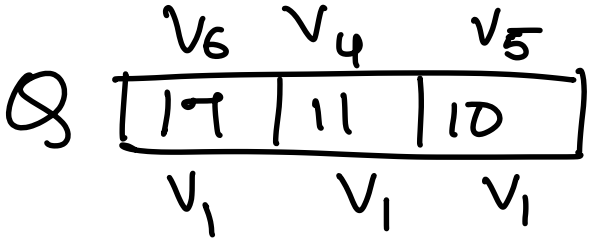
Q

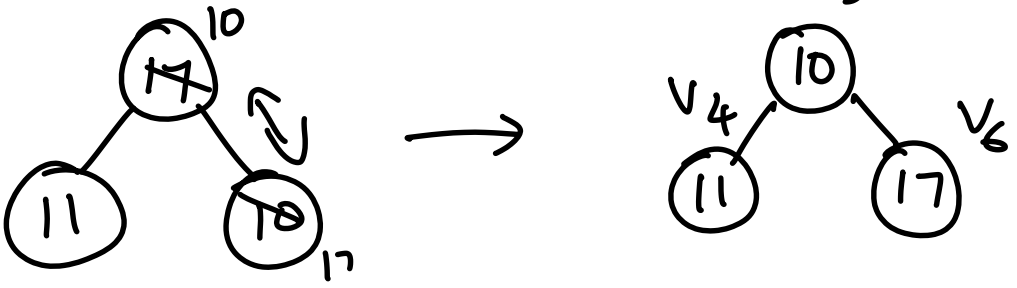
v_3	v_4	v_5	v_6
~~9.5~~ 11	11	10	~~17~~ 9.5
v_2	v_1	v_1	v_1

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Extract Min (Q)

Min ← 9.5





$i = \lfloor \frac{3}{2} \rfloor = 1$

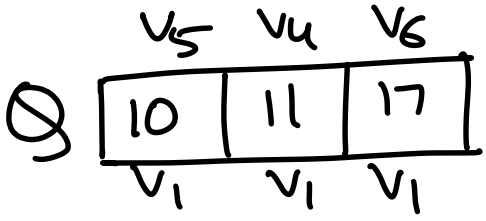
$i=1$

$17 \not< 11$

$11 \not< 10$

$10 \leftrightarrow 17$

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$u \leftarrow v_3$

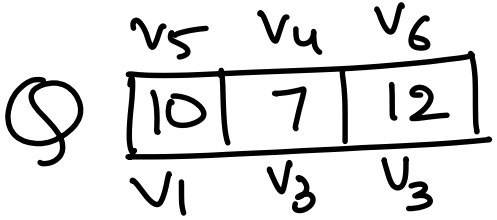
$Adj[v_3] = \{ v_6, v_4, v_1, v_2 \}$ (12, 7, 16, 9.5)

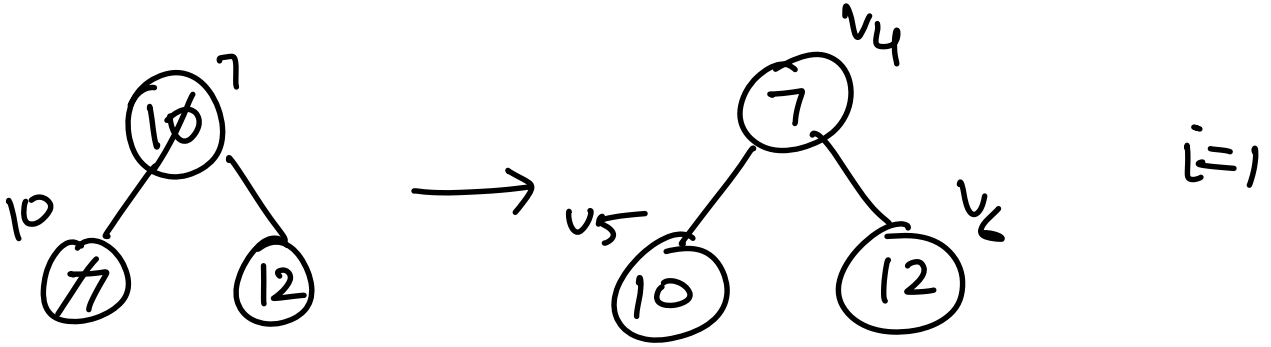
$v_6 \in Q$ and $w(v_3, v_6) = 12 < 17$ update

$v_4 \in Q$ and $w(v_3, v_4) = 7 < 11$ update

$v_1 \notin Q$

$v_2 \notin Q$



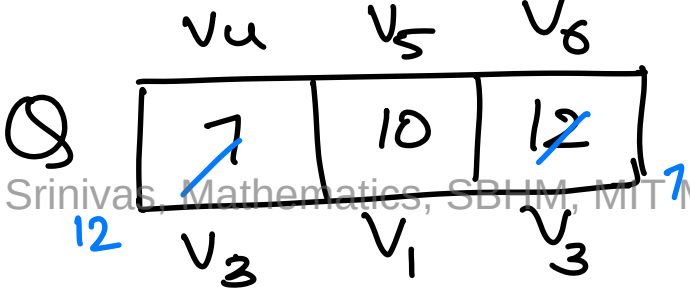


$i=1$

$10 \not< 7$

$7 < 12$

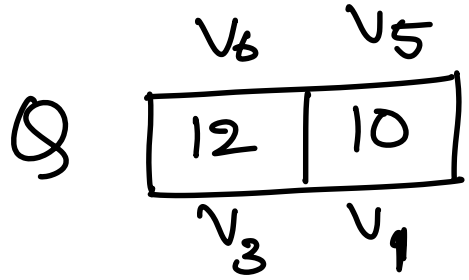
$10 \leftrightarrow 7$

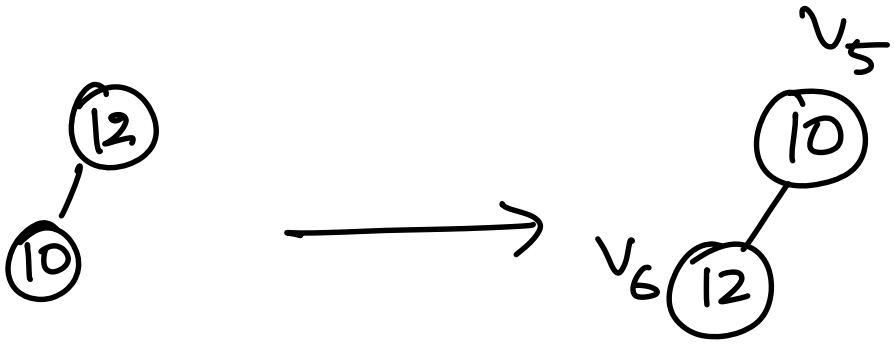


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Extract Min Q.

min ← 7



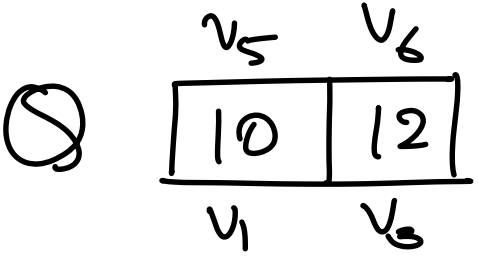


MinHeapify

i=1

12 ≮ 10 ⟺ 10 ≯ 12

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$u \leftarrow v_4$

$Adj[v_4] = \{v_1, v_5, v_6, v_3\}$ (11, 8, 7, 7)

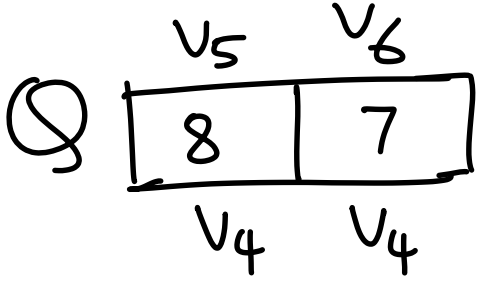
$v_1 \notin Q$

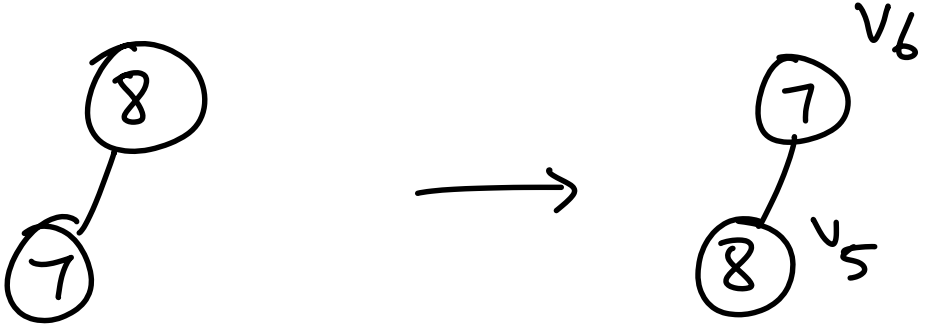
$v_5 \in Q$ and $w(v_4, v_5) = 8 < 10$ update

$v_6 \in Q$ and $w(v_4, v_6) = 7 < 12$ update

$v_3 \notin Q$

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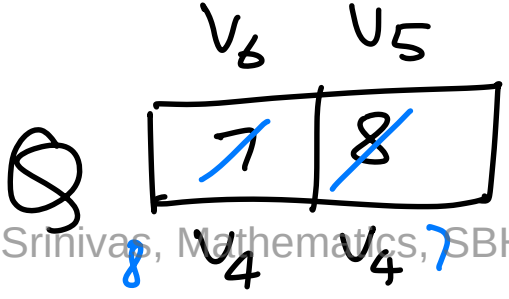




$i=1$

8 ∤ 7

$7 \leftrightarrow 8$.



Extract Min Q.

min ← 7



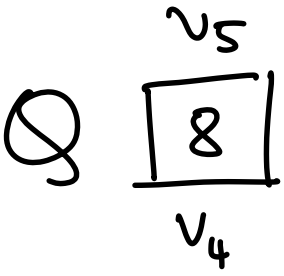
$u \leftarrow v_6$

$Adj[v_6] = \{v_1, v_2, v_3, v_4, v_5\}$ (17, 19.5, 12, 7, 9)

$v_1, v_2, v_3, v_4 \notin Q$, $v_5 \in Q$ $\quad w(v_6, v_5) = 9 \not< 8$

No change

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Valid Heap

Extract Min Q.

$min \leftarrow 8$.

$Q = \phi$.

$u \leftarrow v_5$

$Adj[v_5] = \{ v_1, v_4, v_6 \}$ (10, 8, 9)

$v_1 \notin Q$, $v_4 \notin Q$, $v_6 \notin Q$

$Q = \phi$ (Stop)

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